

Assignment 1 Solutions

Capacity, Double Descent, and Implicit Bias

00103335: Deep Learning and Reinforcement Learning
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These solutions show the main intermediate steps. Empty sums are understood to be zero, and all norms without an explicit subscript in Problem 1 are in $L_2(P_X)$.

Problem 1. Why bias can fall while variance rises

Solution.

Write $\langle g, h \rangle := \mathbb{E}_X[g(X)h(X)]$ and $\|g\|^2 := \langle g, g \rangle$.

- (a) Every $f \in \mathcal{F}_q$ has a unique representation $f = \sum_{j=1}^q a_j \phi_j$. By orthonormality,

$$\|f - m\|^2 = \sum_{j=1}^q (a_j - \theta_j)^2 + \sum_{j=q+1}^Q \theta_j^2.$$

The first sum is minimized uniquely by $a_j = \theta_j$. Hence

$$f_q^* = \sum_{j=1}^q \theta_j \phi_j, \quad \|f_q^* - m\|^2 = \sum_{j=q+1}^Q \theta_j^2.$$

Thus the approximation error is precisely the energy in the omitted coordinates.

- (b) Since $\mathbb{E}_S[Z_j] = \theta_j$,

$$\bar{f}_q = \mathbb{E}_S[\hat{f}_q] = \sum_{j=1}^q \mathbb{E}_S[Z_j] \phi_j = \sum_{j=1}^q \theta_j \phi_j = f_q^*.$$

Consequently, the squared estimator bias is

$$B_q^2 = \left\| \mathbb{E}_S[\hat{f}_q] - m \right\|^2 = \sum_{j=q+1}^Q \theta_j^2.$$

Therefore, for $q < Q$,

$$B_{q+1}^2 - B_q^2 = -\theta_{q+1}^2 \leq 0.$$

Hence B_q^2 is nonincreasing in q , and it decreases strictly at step $q+1$ exactly when $\theta_{q+1} \neq 0$.

- (c) At a fixed x ,

$$\hat{f}_q(x) = \sum_{j=1}^q \theta_j \phi_j(x) + \frac{\nu}{\sqrt{n}} \sum_{j=1}^q \xi_j \phi_j(x).$$

Independence of the ξ_j 's makes the cross-covariances vanish, so

$$\text{Var}_S\{\hat{f}_q(x)\} = \frac{\nu^2}{n} \sum_{j=1}^q \phi_j(x)^2.$$

Integrating and using $\mathbb{E}_X[\phi_j(X)^2] = 1$ gives

$$V_q = \mathbb{E}_X[\text{Var}_S\{\hat{f}_q(X)\}] = \frac{\nu^2}{n} \sum_{j=1}^q \mathbb{E}_X[\phi_j(X)^2] = \frac{q\nu^2}{n}.$$

Therefore, for $q < Q$,

$$V_{q+1} - V_q = \frac{\nu^2}{n}.$$

Since $\nu > 0$, V_q is strictly increasing in q .

Problem 2. Gradient descent selects the minimum-norm interpolator*Solution.*

Let $\mathbf{A} := \mathbf{X}\mathbf{X}^\top$. Since $\mathbf{X}$ has full row rank, $\mathbf{A}$ is symmetric positive definite.

(a) Expanding the objective and differentiating gives

$$\nabla \widehat{L}(b) = -\frac{1}{n} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}b) = \frac{1}{n} \mathbf{X}^\top (\mathbf{X}b - \mathbf{y}).$$

Therefore

$$b_{t+1} = b_t + \frac{\eta}{n} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}b_t).$$

The Hessian is

$$\nabla^2 \widehat{L}(b) = \frac{1}{n} \mathbf{X}^\top \mathbf{X}.$$

Since $\text{rank}(\mathbf{X}) = n < d$, this Hessian has rank n and a nullspace of dimension $d - n$. It is not bounded below by $\mu \mathbf{I}_d$ for any $\mu > 0$, so the theorem for globally strongly convex objectives does not apply on all of $\mathbb{R}^d$.

(b) Put

$$\mathbf{M} := \mathbf{I}_n - \frac{\eta}{n} \mathbf{A}.$$

Using the update,

$$\begin{aligned} r_{t+1} &= \mathbf{y} - \mathbf{X}b_{t+1} \\ &= \mathbf{y} - \mathbf{X}b_t - \frac{\eta}{n} \mathbf{X}\mathbf{X}^\top (\mathbf{y} - \mathbf{X}b_t) \\ &= \mathbf{M}r_t. \end{aligned}$$

Since $r_0 = \mathbf{y}$, we have $r_t = \mathbf{M}^t \mathbf{y}$.

Diagonalize

$$\mathbf{A} = \mathbf{U} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{U}^\top, \quad \lambda_i > 0.$$

Then

$$\mathbf{M}^t = \mathbf{U} \text{diag} \left(\left(1 - \frac{\eta\lambda_1}{n}\right)^t, \dots, \left(1 - \frac{\eta\lambda_n}{n}\right)^t \right) \mathbf{U}^\top.$$

If $\mathbf{u}_i$ denotes column i of $\mathbf{U}$, then

$$\begin{aligned} \|r_t\|_2^2 &= \sum_{i=1}^n \left| 1 - \frac{\eta\lambda_i}{n} \right|^{2t} (\mathbf{u}_i^\top \mathbf{y})^2 \\ &\leq \rho^{2t} \|\mathbf{y}\|_2^2, \quad \rho := \max_i \left| 1 - \frac{\eta\lambda_i}{n} \right|. \end{aligned}$$

All spectral factors have magnitude less than one precisely when

$$0 < \eta < \frac{2n}{\lambda_{\max}(\mathbf{A})} = \frac{2n}{\|\mathbf{X}\|_{\text{op}}^2}.$$

On this interval $\rho < 1$, and therefore $\|r_t\|_2 \leq \rho^t \|\mathbf{y}\|_2 \rightarrow 0$.

(c) Summing the first t updates and using $r_s = \mathbf{M}^s \mathbf{y}$,

$$b_t = \frac{\eta}{n} \mathbf{X}^\top \sum_{s=0}^{t-1} \mathbf{M}^s \mathbf{y}.$$

Since $\mathbf{I}_n - \mathbf{M} = (\eta/n)\mathbf{A}$ is invertible, the matrix geometric series gives

$$\sum_{s=0}^{t-1} \mathbf{M}^s = (\mathbf{I}_n - \mathbf{M}^t)(\mathbf{I}_n - \mathbf{M})^{-1}.$$

Thus

$$\begin{aligned} b_t &= \mathbf{X}^\top (\mathbf{I}_n - \mathbf{M}^t) \mathbf{A}^{-1} \mathbf{y} \\ &= \mathbf{X}^\top \mathbf{A}^{-1} (\mathbf{I}_n - \mathbf{M}^t) \mathbf{y}. \end{aligned}$$

The second equality holds because $\mathbf{M}$ is a polynomial in $\mathbf{A}$, so $\mathbf{M}^t$ commutes with $\mathbf{A}^{-1}$. Consequently,

$$b_t = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \left[\mathbf{I}_n - \left(\mathbf{I}_n - \frac{\eta}{n} \mathbf{X}\mathbf{X}^\top \right)^t \right] \mathbf{y}.$$

Because $\|\mathbf{M}^t\|_{\text{op}} \leq \rho^t \rightarrow 0$,

$$b_t \longrightarrow b^\dagger := \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{y}.$$

(d) Directly,

$$\mathbf{X}b^\dagger = \mathbf{X}\mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{y} = \mathbf{y},$$

so $b^\dagger$ interpolates. If b is any interpolator, then $\mathbf{X}(b - b^\dagger) = 0$. Hence

$$b = b^\dagger + v, \quad v \in \ker(\mathbf{X}).$$

But $b^\dagger \in \text{range}(\mathbf{X}^\top) = \ker(\mathbf{X})^\perp$, so $\langle b^\dagger, v \rangle = 0$. Pythagoras yields

$$\|b\|_2^2 = \|b^\dagger\|_2^2 + \|v\|_2^2 \geq \|b^\dagger\|_2^2,$$

with equality only for $v = 0$. Thus $b^\dagger$ is the unique minimum-norm interpolator.

(e) For arbitrary b_0 , the residual recursion becomes

$$r_t = \mathbf{M}^t r_0, \quad r_0 = \mathbf{y} - \mathbf{X}b_0.$$

Summing the updates now gives

$$b_t = b_0 + \mathbf{X}^\top \mathbf{A}^{-1} (\mathbf{I}_n - \mathbf{M}^t) (\mathbf{y} - \mathbf{X}b_0).$$

Under the same step-size condition,

$$\begin{aligned} b_t &\longrightarrow b_0 + \mathbf{X}^\top \mathbf{A}^{-1} (\mathbf{y} - \mathbf{X}b_0) \\ &= b^\dagger + [\mathbf{I}_d - \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}] b_0. \end{aligned}$$

The matrix

$$\mathbf{P} := \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}$$

is symmetric, idempotent, and has range $\text{range}(\mathbf{X}^\top)$. It is therefore the orthogonal projector onto $\ker(\mathbf{X})^\perp$, and $\mathbf{I}_d - \mathbf{P} = \Pi_{\ker(\mathbf{X})}$. Hence

$$b_t \longrightarrow b^\dagger + \Pi_{\ker(\mathbf{X})} b_0.$$

GD updates can alter only the row-space component, so the initial nullspace component is preserved. Zero initialization makes that component zero. More generally, any b_0 with $\Pi_{\ker(\mathbf{X})} b_0 = 0$ also converges to the minimum-norm interpolator.

Problem 3. Hat matrices, residuals, and interpolation

Solution.

Throughout parts (a)–(e), $d < n$ and $\mathbf{X}$ has full column rank.

- (a) Since $\mathbf{X}^\top \mathbf{X}$ is symmetric positive definite, its inverse is symmetric. Therefore

$$\mathbf{H}^\top = \mathbf{X}\{(\mathbf{X}^\top \mathbf{X})^{-1}\}^\top \mathbf{X}^\top = \mathbf{H}.$$

Also,

$$\begin{aligned}\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}.\end{aligned}$$

- (b) If $z \in \text{range}(\mathbf{X})$, write $z = \mathbf{X}a$. Then

$$\mathbf{H}z = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}a = \mathbf{X}a = z.$$

If $z \in \ker(\mathbf{X}^\top)$, then $\mathbf{H}z = 0$. Conversely, $\mathbf{H}z = 0$ implies

$$\mathbf{X}^\top z = \mathbf{X}^\top \mathbf{H}z = 0,$$

where $\mathbf{X}^\top \mathbf{H} = \mathbf{X}^\top$. Thus

$$\text{range}(\mathbf{H}) = \text{range}(\mathbf{X}), \quad \ker(\mathbf{H}) = \ker(\mathbf{X}^\top).$$

A symmetric idempotent matrix is an orthogonal projector, so $\mathbf{H}$ projects onto $\text{range}(\mathbf{X})$. Its complementary projector $\mathbf{I}_n - \mathbf{H}$ projects onto $\ker(\mathbf{X}^\top) = \text{range}(\mathbf{X})^\perp$.

- (c) If $\mathbf{H}v = \lambda v$ and $v \neq 0$, idempotence gives $\lambda^2 v = \lambda v$, so $\lambda \in \{0, 1\}$. The one-eigenspace is $\text{range}(\mathbf{X})$, of dimension d , and the zero-eigenspace is $\ker(\mathbf{X}^\top)$, of dimension $n - d$. Hence

$$\text{rank}(\mathbf{H}) = \text{tr}(\mathbf{H}) = d, \quad \text{tr}(\mathbf{I}_n - \mathbf{H}) = n - d.$$

For the diagonal entries,

$$h_{ii} = e_i^\top \mathbf{H} e_i = e_i^\top \mathbf{H}^\top \mathbf{H} e_i = \|\mathbf{H} e_i\|_2^2 \geq 0.$$

Orthogonal projection cannot increase norm, so $\|\mathbf{H} e_i\|_2 \leq \|e_i\|_2 = 1$, which gives $h_{ii} \leq 1$. Finally, $\sum_i h_{ii} = \text{tr}(\mathbf{H}) = d$.

- (d) The least-squares normal equations give

$$\hat{b} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Therefore

$$\hat{\mathbf{y}} = \mathbf{X} \hat{b} = \mathbf{H} \mathbf{y}, \quad r = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H}) \mathbf{y}.$$

Since $\mathbf{X}^\top \mathbf{H} = \mathbf{X}^\top$,

$$\mathbf{X}^\top r = \mathbf{X}^\top (\mathbf{I}_n - \mathbf{H}) \mathbf{y} = 0.$$

The two terms in $\mathbf{y} = \mathbf{H} \mathbf{y} + (\mathbf{I}_n - \mathbf{H}) \mathbf{y}$ lie in orthogonal subspaces. Hence

$$\|\mathbf{y}\|_2^2 = \|\mathbf{H} \mathbf{y}\|_2^2 + \|(\mathbf{I}_n - \mathbf{H}) \mathbf{y}\|_2^2.$$

The minimized empirical loss is therefore

$$\min_b \frac{1}{2n} \|\mathbf{y} - \mathbf{X}b\|_2^2 = \frac{1}{2n} \|(\mathbf{I}_n - \mathbf{H}) \mathbf{y}\|_2^2.$$

(e) Because $\mathbf{H}$ is the identity on $\text{range}(\mathbf{X})$,

$$(\mathbf{I}_n - \mathbf{H})\mathbf{X}\beta = 0.$$

It follows that $(\mathbf{I}_n - \mathbf{H})\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\eta$. Conditional on $\mathbf{X}$,

$$\begin{aligned} \mathbb{E}[\|(\mathbf{I}_n - \mathbf{H})\mathbf{y}\|_2^2 \mid \mathbf{X}] &= \mathbb{E}[\eta^\top (\mathbf{I}_n - \mathbf{H})^\top (\mathbf{I}_n - \mathbf{H})\eta \mid \mathbf{X}] \\ &= \mathbb{E}[\eta^\top (\mathbf{I}_n - \mathbf{H})\eta \mid \mathbf{X}] \\ &= \text{tr}((\mathbf{I}_n - \mathbf{H})\mathbb{E}[\eta\eta^\top \mid \mathbf{X}]) \\ &= \tau^2 \text{tr}(\mathbf{I}_n - \mathbf{H}) = \tau^2(n - d). \end{aligned}$$

Problem 4. Inverse moment of a Gaussian Gram matrix

Solution.

Let $\mathbf{Z} = [z_1, \dots, z_p] \in \mathbb{R}^{m \times p}$ have independent standard Gaussian entries and $\mathbf{W} = \mathbf{Z}^\top \mathbf{Z}$.

(a) Since $m > p + 1$, in particular $p < m$. Expose the columns of $\mathbf{Z}$ successively. Conditional on the preceding columns, their span is a fixed proper subspace of $\mathbb{R}^m$, and the next independent Gaussian column belongs to that subspace with probability zero. Hence $\mathbf{Z}$ has full column rank almost surely. For every nonzero $v \in \mathbb{R}^p$,

$$v^\top \mathbf{W} v = \|\mathbf{Z}v\|_2^2 > 0$$

almost surely. Hence $\mathbf{W}$ is positive definite almost surely.

(b) If $p = 1$, then $\mathbf{W} = [z_1^\top z_1]$ and, by convention, $\mathbf{P}_{-1} = 0$; the claimed identity follows directly. For $p \geq 2$, by part (a), $\mathbf{Z}_{-j}$ has full column rank $p - 1$ almost surely, so $\mathbf{P}_{-j}$ is the orthogonal projector onto $\text{range}(\mathbf{Z}_{-j})$ and has rank $p - 1$. Permute the columns so that z_j is first and write

$$\widetilde{\mathbf{W}} = \begin{pmatrix} a & b^\top \\ b & \mathbf{C} \end{pmatrix}, \quad a = z_j^\top z_j, \quad b = \mathbf{Z}_{-j}^\top z_j, \quad \mathbf{C} = \mathbf{Z}_{-j}^\top \mathbf{Z}_{-j}.$$

The scalar Schur complement of $\mathbf{C}$ is

$$\begin{aligned} s &= a - b^\top \mathbf{C}^{-1} b \\ &= z_j^\top [\mathbf{I}_m - \mathbf{Z}_{-j}(\mathbf{Z}_{-j}^\top \mathbf{Z}_{-j})^{-1} \mathbf{Z}_{-j}^\top] z_j \\ &= z_j^\top (\mathbf{I}_m - \mathbf{P}_{-j}) z_j. \end{aligned}$$

The block inverse formula gives $[\widetilde{\mathbf{W}}^{-1}]_{11} = s^{-1}$. Undoing the permutation yields

$$\{[\mathbf{W}^{-1}]_{jj}\}^{-1} = z_j^\top (\mathbf{I}_m - \mathbf{P}_{-j}) z_j.$$

(c) Conditional on $\mathbf{Z}_{-j}$, the vector z_j remains $\mathcal{N}(0, \mathbf{I}_m)$, while $\mathbf{I}_m - \mathbf{P}_{-j}$ is an orthogonal projector of rank

$$k = m - (p - 1) = m - p + 1.$$

Choose an orthogonal $\mathbf{U}$ that diagonalizes this projector:

$$\mathbf{U}^\top (\mathbf{I}_m - \mathbf{P}_{-j}) \mathbf{U} = \text{diag}(\mathbf{I}_k, \mathbf{0}).$$

Conditionally, $g := \mathbf{U}^\top z_j \sim \mathcal{N}(0, \mathbf{I}_m)$. Therefore

$$z_j^\top (\mathbf{I}_m - \mathbf{P}_{-j}) z_j = \sum_{\ell=1}^k g_\ell^2 \sim \chi_k^2, \quad k = m - p + 1.$$

(d) If $U \sim \chi_k^2$, then, by its density,

$$\mathbb{E}[U^{-1}] = \frac{1}{2^{k/2}\Gamma(k/2)} \int_0^\infty u^{k/2-2} e^{-u/2} du.$$

For $k > 2$, set $\alpha = k/2 - 1 > 0$. The Gamma integral and recurrence give

$$\begin{aligned} \mathbb{E}[U^{-1}] &= \frac{2^\alpha \Gamma(\alpha)}{2^{k/2} \Gamma(k/2)} = \frac{1}{2} \frac{\Gamma(k/2 - 1)}{\Gamma(k/2)} \\ &= \frac{1}{k - 2}. \end{aligned}$$

If $k \leq 2$, the exponent $k/2 - 2 \leq -1$, and the integral diverges at zero. Hence

$$\mathbb{E}[U^{-1}] = \begin{cases} 1/(k - 2), & k > 2, \\ +\infty, & k \leq 2. \end{cases}$$

Here $k = m - p + 1 > 2$. Parts (b)–(c) and iterated expectation yield

$$\mathbb{E}[(\mathbf{W}^{-1})_{jj}] = \frac{1}{m - p - 1}.$$

(e) First we verify that the off-diagonal entries are integrable. Since $\mathbf{W}^{-1} \succ 0$, each 2×2 principal minor is nonnegative, and hence

$$|[\mathbf{W}^{-1}]_{rs}|^2 \leq [\mathbf{W}^{-1}]_{rr} [\mathbf{W}^{-1}]_{ss}.$$

By Cauchy–Schwarz and part (d),

$$\begin{aligned} \mathbb{E}[|[\mathbf{W}^{-1}]_{rs}|] &\leq \mathbb{E}\left[\sqrt{[\mathbf{W}^{-1}]_{rr} [\mathbf{W}^{-1}]_{ss}}\right] \\ &\leq \sqrt{\mathbb{E}[\mathbf{W}^{-1}]_{rr} \mathbb{E}[\mathbf{W}^{-1}]_{ss}} = \frac{1}{m - p - 1} < \infty. \end{aligned}$$

Fix $r \neq s$, and let $\mathbf{D}$ be diagonal with $D_{rr} = -1$ and all other diagonal entries equal to one. Flipping column r of $\mathbf{Z}$ does not change its distribution, so

$$\mathbf{D}\mathbf{W}^{-1}\mathbf{D} \stackrel{d}{=} \mathbf{W}^{-1}.$$

The (r, s) entry on the left is $-[\mathbf{W}^{-1}]_{rs}$. Integrability therefore gives

$$\mathbb{E}[[\mathbf{W}^{-1}]_{rs}] = -\mathbb{E}[[\mathbf{W}^{-1}]_{rs}] = 0.$$

Combining this with the diagonal value from part (d) gives

$$\boxed{\mathbb{E}[\mathbf{W}^{-1}] = \frac{\mathbf{I}_p}{m - p - 1}}.$$